

ANAGHA MV SIMHA

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RESEARCH INTERESTS

Machine Learning; Natural Language Processing; Domain Adaptation under Distribution Shift; Causal Inference Methods

EDUCATION

MS in Engineering Data Analytics and Statistics

May 2025

Washington University in St. Louis, Dept. of Electrical & Systems Engineering

GPA: 3.84 / 4.00 | Graduate Certificate in Financial Engineering | Ascend Global Leaders Network

BE in Electronics and Communication Engineering

June 2022

University Visvesvaraya College of Engineering (UVCE), Bengaluru

CGPA: 8.86 / 10.00

WORKING PAPERS

Simha, A., Pagel, M. "Domain-Adapted Text Classification and Causal Estimation under Distribution Shift in Online Communities."

Status: Empirical analysis and ML pipeline finalized; manuscript draft in progress, targeting completion Fall 2026. Draft available [here](#).

Simha, A., Yimam, S.M., et al. "Proactive Hate Speech Detection Around Major Events." In preparation.

Status: Literature review in progress. Lead author. Collaboration with University of Hamburg and IIT Kharagpur.

RESEARCH EXPERIENCE

Washington University in St. Louis, Olin School of Business

St. Louis, USA (Remote)

Research Assistant Intern

September 2025 – Present

Principal Investigator: Dr. Michaela Pagel

- Designed a quasi-experimental framework combining NLP classification with causal estimation to measure treatment effects of external shocks on user posting behavior across fourteen online communities (36M+ posts).
- Built chunked PostgreSQL ingestion pipelines and a longitudinal user-week panel to process 25GB+ of unstructured forum data.
- Fine-tuned a DistilBERT sequence classifier for 3-class text classification on a fuzzy social construct.
- Constructed multi-decade treatment variables from BLS releases and Survey of Professional Forecasters data for use in causal estimation pipelines.
- Built a 19-event intervention log (2015–2021) from news archives and academic literature.
- Conducted three ablation experiments (DeBERTa architecture comparison, LoRA rank sweep $r \in \{4, 8, 16, 32\}$, and leave-one-forum-out evaluation) to diagnose performance ceiling.

Washington University in St. Louis, Dept. of Electrical & Systems Engineering

St. Louis, USA

Graduate Student Researcher

January 2024 – May 2024

Principal Investigator: Dr. Jr Shin-Li | Direct Supervisor: Bharat Singhal (PhD Candidate)

- Formulated and analyzed oscillatory models quantifying non-stationary noise characteristics and inter-signal phase variability in coupled dynamical systems.
- Implemented and extended coupled phase-oscillator frameworks (Kuramoto-type dynamics) in Python and MATLAB to evaluate complex network synchronization boundaries under varying stochastic noise regimes.
- Investigated stability boundaries and collective phase dynamics across parameterized simulation pipelines; contributed findings to the lab's broader network synchronization research agenda.

University Visvesvaraya College of Engineering (UVCE)

Bengaluru, India

Undergraduate Research Intern June 2021 – August 2021
Principal Investigator: Dr. Janamejaya Channegowda

- Designed and implemented a federated learning pipeline for battery performance prediction.
- Conducted literature review and reproduced key experiments from prior work.

WORK EXPERIENCE

International Career Advisory (ICAWay) Remote
Machine Learning Intern August 2025 – Present

- Built an end-to-end LLM chatbot pipeline using Ollama over curated educational content; deployed and hosted on AWS.

KPMG India Bengaluru, India
Analyst July 2022 – June 2023

- Supported financial systems implementation for US banking clients using Salesforce; contributed to data migration and client delivery workflows.

Elektrobit India Bengaluru, India
Machine Learning Intern January 2022 – April 2022

- Automated a manual hand-labeling process using ML algorithms; worked with sensitive automotive software requirements documents under strict data handling protocols.

TEACHING EXPERIENCE

Washington University in St. Louis, Dept. of Electrical & Systems Engineering. St. Louis, USA
Graduate Teaching Assistant – Dr Vladimir Kurenok Spring 2025

- Conducted regular instructional office hours and review sessions to clarify complex graduate-level statistical methods, probability concepts, and data analytics principles for student cohorts.
- Designed rigorous, structured evaluation criteria for graduate-level assignments and examinations, translating abstract academic benchmarks into transparent and measurable metrics of performance.

TECHNICAL SKILLS

Languages: Python, SQL, MATLAB, R, C++

Methods and Models: Causal inference, event-study design, pre-trend validation, panel data methods, time-series alignment, Transformer architectures(BERT, DistilBERT)

Frameworks and Tools: PyTorch, Hugging Face Transformers, Pandas, NumPy, SciPy, Scikit-learn, XGBoost

RELEVANT COURSEWORK

Optimization | Probability and Stochastic Processes | Introduction to Machine Learning | Practicum in Data Analytics and Statistics | Financial Engineering | Statistics for Finance | Quantitative Risk Management